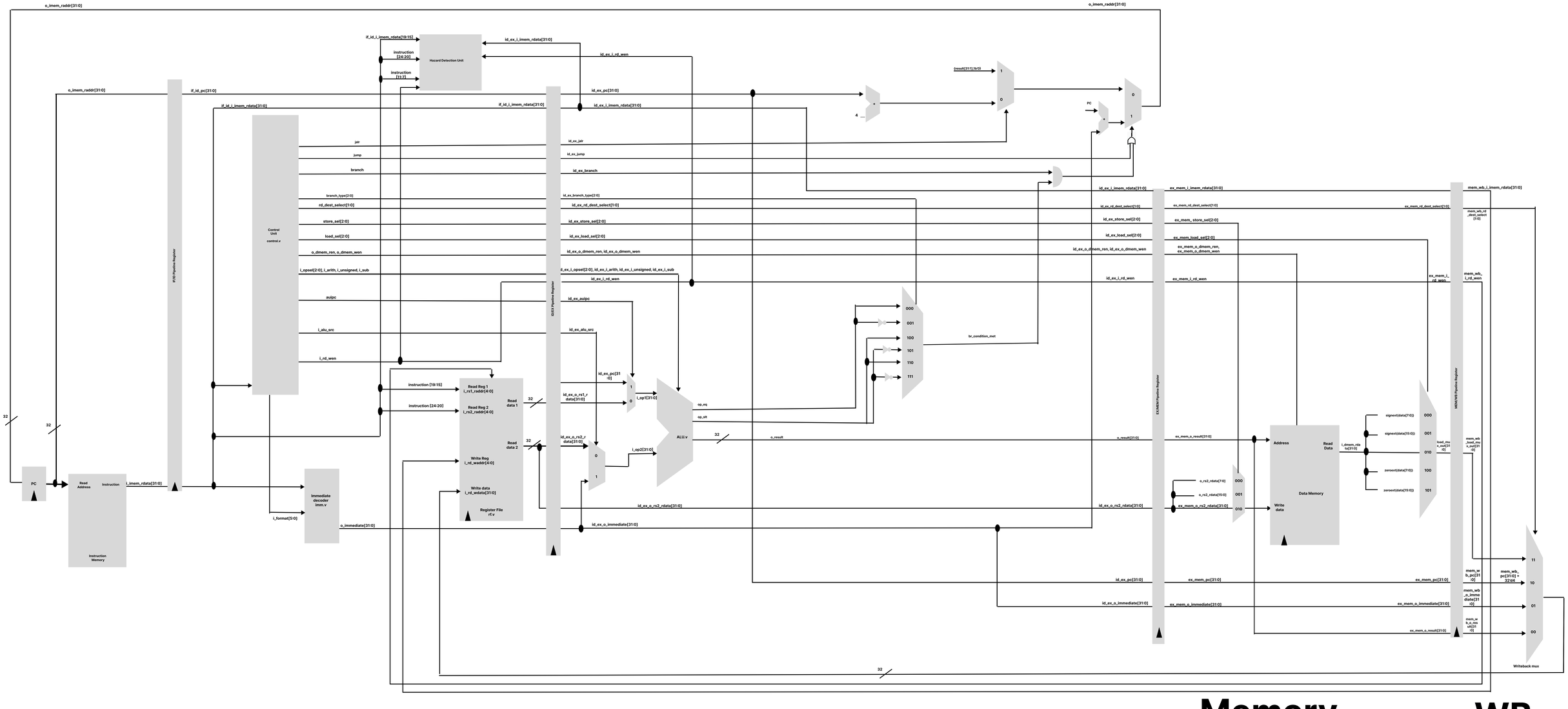
o_imem_raddr[31:0]
o_imem_raddr[31:0]
if_id_i_imem_rdata[19:15]
instruction [24:20]
instruction [11:7]
Hazard Detection Unit
id_ex_i_imem_rdata[31:0]
id_ex_i_rd_wen
o_imem_raddr[31:0]
if_id_pc[31:0]
id_ex_pc[31:0]
(result[31:1],1'b'0)
1
0
4
+
PC
+
0
1
if_id_i_imem_rdata[31:0]
if_id_i_imem_rdata[31:0]
id_ex_i_imem_rdata[31:0]
IF/ID Pipeline Register
Control Unit
control.v
jalr
jump
branch
branch_type[2:0]
rd_dest_select[1:0]
store_sel[2:0]
load_sel[2:0]
o_dmem_ren, o_dmem_wen
i_opsel[2:0], i_arith, i_unsigned, i_sub
auipc
i_alu_src
i_rd_wen
id_ex_jalr
id_ex_jump
id_ex_branch
id_ex_branch_type[2:0]
id_ex_rd_dest_select[1:0]
id_ex_store_sel[2:0]
id_ex_load_sel[2:0]
id_ex_o_dmem_ren, id_ex_o_dmem_wen
id_ex_i_opsel[2:0], id_ex_i_arith, id_ex_i_unsigned, id_ex_i_sub
id_ex_i_rd_wen
id_ex_auipc
id_ex_alu_src
id_ex_pc[31:0]
id_ex_o_rs1_rdata[31:0]
id_ex_o_rs2_rdata[31:0]
id_ex_o_immediate[31:0]
ID/EX Pipeline Register
000
001
100
101
110
111
br_condition_met
32
32
PC
Read Address
Instruction
i_imem_rdata[31:0]
Instruction Memory
instruction [19:15]
instruction [24:20]
Read Reg 1
i_rs1_raddr[4:0]
Read Reg 2
i_rs2_raddr[4:0]
Write Reg
i_rd_waddr[4:0]
Write data
i_rd_wdata[31:0]
Read data 1
Read data 2
Register File
rf.v
32
32
Immediate decoder
imm.v
i_format[5:0]
o_immediate[31:0]
1
0
i_op1[31:0]
0
1
i_op2[31:0]
ALU.v
op_eq
op_slt
32
o_result
id_ex_o_rs2_rdata[31:0]
id_ex_o_immediate[31:0]
id_ex_i_imem_rdata[31:0]
id_ex_rd_dest_select[1:0]
id_ex_store_sel[2:0]
id_ex_load_sel[2:0]
id_ex_o_dmem_ren, id_ex_o_dmem_wen
id_ex_i_rd_wen
o_result[31:0]
id_ex_o_rs2_rdata[31:0]
id_ex_pc[31:0]
id_ex_o_immediate[31:0]
EX/MEM Pipeline Register
ex_mem_i_imem_rdata[31:0]
ex_mem_rd_dest_select[1:0]
ex_mem_ store_sel[2:0]
ex_mem_load_sel[2:0]
ex_mem_o_dmem_ren,
ex_mem_o_dmem_wen
ex_mem_i_rd_wen
ex_mem_o_result[31:0]
o_rs2_rdata[7:0]
o_rs2_rdata[15:0]
ex_mem_o_rs2_rdata[31:0]
000
001
010
Address
Read Data
Data Memory
Write data
i_dmem_rdata[31:0]
signext(data[7:0])
signext(data[15:0])
zeroext(data[7:0])
zeroext(data[15:0])
000
001
010
100
101
ex_mem_pc[31:0]
ex_mem_o_immediate[31:0]
ex_mem_o_result[31:0]
MEM/WB Pipeline Register
mem_wb_i_imem_rdata[31:0]
ex_mem_rd_dest_select[1:0]
mem_wb_rd_dest_select[1:0]
ex_mem_i_rd_wen
mem_wb_i_rd_wen
mem_wb_load_mux_sel[31:0]
mem_wb_pc[31:0]
mem_wb_o_immediate[31:0]
mem_wb_o_result[31:0]
mem_wb_pc[31:0] + 32'd4
11
10
01
00
Writeback mux
32
Fetch
Decode
Execute
Memory
WB